- Inductive biases are the prior information we have about some set of data; we use them to constrain our model toward learning certain kinds of patterns. In this chapter we employed a relational inductive bias, for example.
- We say a function, *f* is invariant to some transformation, *g*, when the function's output remains unchanged when the input is first transformed by *g*: $f(g(x)) = f(x)$.
- We say a function, *f*, is equivariant to some transformation, *g*, when applying the transformation to the input is the same as applying the transformation to the output: $f(g(x)) = g(f(x))$.
- Attention models are designed to increase the interpretability and performance of machine learning models by forcing the model to only "look at" (attend to) a subset of the input data. By examining what the model learns to attend to, we can have a better idea of how it is making decisions.
- Self-attention models model attention between objects (or nodes) in an input rather than just the model attending to different parts of the input. This naturally leads to a form of graph neural network, since the attention weights can be interpreted as edges between nodes.
- Multi-head self-attention allows the model to have independent attention mechanisms that can each attend to a different subset of the input data. This allows us to still get interpretable attention weights but increases the bandwidth of information that can flow through the model.
- Relational reasoning is a form of reasoning based on objects and relationships between objects, rather than using absolute frames of reference. For example, "a book is on the top" relates the book to the table, rather than saying the book is at position 10 and the table is at position 8 (an absolute reference frame).
- Inner (dot) product is a product between two vectors that results in a single scalar value.
- Outer product is a product between two vectors that results in a matrix.
- Einstein notation or Einsum lets us describe generalized tensor-tensor products called tensor contractions using a simple syntax based on labeling the indices of a tensor.
- Double Q-learning stabilizes training by separating action-selection and action-value updating.